

Cubic Function Report

Polynomial

$$f(q) = aq^3 + bq^2 + cq + d$$

Naive

$$g_{\text{naive}}(X) = X^3a + X^2b + Xc + d$$

$$\mathbb{E}[g_{\text{naive}}(X)] = aq^3 + bq^2 + 2\frac{\Delta^2}{\epsilon}(3aq + b) + cq + d$$

$$\begin{aligned} \text{Var}(g_{\text{naive}}(X)) = 2\frac{\Delta^2}{\epsilon} & \left(360a^2\frac{\Delta^4}{\epsilon} + 9a^2q^4 + 12abq^3 + 6acq^2 + 4b^2q^2 + 4bcq \right. \\ & \left. + 2\frac{\Delta^2}{\epsilon} (81a^2q^2 + 54abq + 12ac + 5b^2) + c^2 \right) \end{aligned}$$

$$\begin{aligned} \text{MSE}(g_{\text{naive}}) = 2\frac{\Delta^2}{\epsilon} & \left(360a^2\frac{\Delta^4}{\epsilon} + 9a^2q^4 + 12abq^3 + 6acq^2 + 4b^2q^2 + 4bcq \right. \\ & \left. + 12\frac{\Delta^2}{\epsilon} (15a^2q^2 + 10abq + 2ac + b^2) + c^2 \right) \end{aligned}$$

$$\text{Bias}(g_{\text{naive}}) = 2\frac{\Delta^2}{\epsilon}(3aq + b)$$

Unbiased

$$g_{\text{unb}}(X) = X^3a + X^2b + Xc - 2\frac{\Delta^2}{\epsilon}(3Xa + b) + d$$

$$\mathbb{E}[g_{\text{unb}}(X)] = aq^3 + bq^2 + cq + d$$

$$\begin{aligned} \text{Var}(g_{\text{unb}}(X)) = 2\frac{\Delta^2}{\epsilon} & \left(252a^2\frac{\Delta^4}{\epsilon} + 9a^2q^4 + 12abq^3 + 6acq^2 + 4b^2q^2 + 4bcq \right. \\ & \left. + 2\frac{\Delta^2}{\epsilon} (63a^2q^2 + 42abq + 6ac + 5b^2) + c^2 \right) \end{aligned}$$

$$\begin{aligned} \text{MSE}(g_{\text{unb}}) = 2\frac{\Delta^2}{\epsilon} & \left(252a^2\frac{\Delta^4}{\epsilon} + 9a^2q^4 + 12abq^3 + 6acq^2 + 4b^2q^2 + 4bcq \right. \\ & \left. + 2\frac{\Delta^2}{\epsilon} (63a^2q^2 + 42abq + 6ac + 5b^2) + c^2 \right) \end{aligned}$$

Comparison

$$\text{mean gap} = 2 \frac{\Delta^2}{\epsilon} (-3aq - b)$$

$$\text{variance gap} = -24a \frac{\Delta^4}{\epsilon} \left(9a \frac{\Delta^2}{\epsilon} + 3aq^2 + 2bq + c \right)$$

$$\text{MSE gap} = 4 \frac{\Delta^4}{\epsilon} \left(-54a^2 \frac{\Delta^2}{\epsilon} - 27a^2 q^2 - 18abq - 6ac - b^2 \right)$$

$$\text{Bias}(g_{\text{naive}})^2 = 4 \frac{\Delta^4}{\epsilon} (9a^2 q^2 + 6abq + b^2)$$